

Title

Password Strength Checker: A Cyber Security Tool for Secure Password Evaluation

Project Name

Development of a Password Strength Checker Using Python

Objective

The main objective of this project is to develop a Password Strength Checker that evaluates the strength of a user's password. The program helps users create secure passwords by checking the length and the use of uppercase letters, lowercase letters, numbers, and special characters. This improves account security and reduces the risk of cyber attacks.

Problem Statement

Many users create weak passwords that are easy to guess or crack, making their personal information and online accounts vulnerable to cyber attacks. The lack of awareness about strong password practices increases the risk of unauthorized access. This project aims to solve this problem by checking password strength and encouraging users to create secure password.

Research Findings

During the research, it was found that weak passwords are one of the main causes of cyber attacks and unauthorized access. A strong password should contain at least 8 characters, including uppercase letters, lowercase letters, numbers, and special characters. Using strong and unique passwords helps protect personal information and improves online security.

Methodology

The project was developed using Python. The program takes a password as input and checks its length and the presence of uppercase letters, lowercase letters, numbers, and special characters. Based on these checks, it classifies the password as Weak, Medium, or Strong.

Tools Used

- Python
- Visual Studio Code (VS Code)
- Windows 10/11

Implementation

The Password Strength Checker was developed using Python. The program accepts a password from the user and checks its length, uppercase letters, lowercase letters, numbers, and special characters. Based on these checks, it classifies the password as Weak, Medium, or Strong.

Source Code

```
Password = input("Enter your password: ")

Length = len(password) >= 8

Upper = any(ch.isupper() for ch in password)

Lower = any(ch.islower() for ch in password)

Digit = any(ch.isdigit() for ch in password)

Special = any(not ch.isalnum() for ch in password)

Score = length + upper + lower + digit + special

If score == 5:
    _Print("Strong Password")

Elif score >= 3:
    _Print("Medium Password")

Else:
    _Print("Weak Password")
```

Screenshot



TAB



Enter your password: Hello@123
Strong Password

[Program finished]

Results

The program was tested successfully. It correctly identified “Hello@123” as a Strong Password. The password met all security requirements.

Challenges

Understanding password validation logic.

Running the Python code on a mobile phone.

Formatting the source code in the document.

Future Scope

Add password suggestions.

Display password strength using colors or a progress bar.

Check for commonly used weak passwords.

Conclusion

The Password Strength Checker is a simple and useful Python project. It helps users create stronger passwords and improves cybersecurity awareness.